ERIC HOU

hyweric.github.io | hou239@purdue.edu | +1-510-288-0541

EDUCATION

PURDUE UNIVERSITY, West Lafayette, IN — *Present*

- Bachelor of Science | College of Engineering

MISSION SAN JOSE HIGH SCHOOL, Fremont, CA — *GPA 4.38 Weighted*

- Coursework: Multivariable Calculus, Linear Algebra, Statistics, Computer Science A, Physics Mechanics + E&M
- SAT 1550/1600

RESEARCH EXPERIENCE

IEEE Paper 1st Author — *Sep 2023 - Dec 2024*

- Published at IRC 2024; Developed Quadrupedal Robot w/ Adaptive Suction & Suspension Foot Design for Infrastructure Monitoring

ASDRP Student Researcher — *Aug 2021 - Jun 2023*

- Applied Quantum Annealing for Isolation of Particle Collision Jets Using Regression Models with Dr. McMahan

Embedded Systems | SlayterHIL — *Present*
Hardware / EE Team

- Prototyping Communications PCB to Interface Between DUT and Test Node, Enabling Hardware-in-the-Loop Testing for Drone Flight Controller

PROJECTS

5 Stage Electromagnetic Coil Launcher — *Jan 2025 - May 2025*

Autodesk Fusion 360 Wire Generator Plugin (3000+ Users) — *Oct 2023 - Feb 2024*

Optical Flow Egomotion Localization Algorithm using Computer Vision — *Dec 2023 - Jan 2024*

2DOF / 3DOF Inverse Kinematic Simulator for Robot Arms — *Oct 2023*

Additional Information: hyweric.github.io/projects

LEADERSHIP

FTC TEAM 13356 — *Aug 2021 - Jun 2025*
Hardware Co-captain & Outreach

- Leadership: Oversaw End-to-End Hardware Design, Manufacturing, and Testing for Each Season
- Achievements: California #3 in Performance (Top 30 Worldwide); Qualified for MTI (Top 0.5% Globally) Three Consecutive Years; 2× NorCal Finalist Alliance Captain
- Outreach: Led STEM Kit Initiative - Designed and Produced Hundreds of Custom STEM Kits for Global Organizations; Developed and Taught CAD/Programming Curriculum to Afghan Refugees, Taiwanese Schools, and Rookie Robotics Teams; Organized 8+ Community Events and Managed Outreach Booths at Local Fairs

LEARNIFY+ Nonprofit — *Sep 2021 - Jan 2024*
Head of Computer Science & Board of Director

- Volunteered Over 650 Hours Creating Plans and Organizing Classes for 50+ Tutors and 200+ Students
- Led the Mission International Initiative, Expanding the Organization's Reach to 10 States and 4 Countries
- Created Course Curricula for Java, Python, and HTML for Computer Science Tutors

MSJ High School Robotics Club — *Aug 2022 - Jun 2025*
Club Officer (Various Positions)

- Interviewed Candidates and Distributed Club Roles; Hosted Club Meetings; Planned Outreach Events and Fundraisers

OTHER

SASA (Student Association for Stem Advocacy) — *Jan 2023 - Jan 2025*
Logistics Manager / Lobbyist

- Coordinated Advocacy Campaigns/Conferences; Lobbied to Senators & Congressional Offices

Horizon AI Startup — *Jan 2024 - Jan 2025*
Intern

- Organized User Database on Excel & Improved Website SEO

Toshiba ExploraVision Project | HM Award — *Jan 2023 - Mar 2023*

- Developed Proposal for Eco-friendly Air Purifying System Using Algae PBR Modules

IWUF World Championship — *May 2019 - Aug 2024*
USA International Team Member

- Represented the U.S. in WWC (World Wushu Championships)

PSP Active Controls Avionics Team — *Present*

SKILLS

Design / Manufacturing: KiCAD (Schematic / PCB Design), FDM-3D Printing, Waterjet, CNC, Fusion 360 CAD/CAM/FEA

Languages: Python, Java, C, HTML/JS/CSS